Jacob Hanson-Regalado

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EXPERIENCE

Adobe Digital Experience

Machine Learning Engineer | May 2021 – August 2021, August 2022 - present

- Lead a team of 5 engineers developing an AI product for translating non-technical marketing queries into domain-specific code, combining modern generative AI, and classical statistical techniques.
- Build data mining pipelines to support AI workflows using PySpark.
- Manage multiple backend services for AI model inference and access to resources stored in MongoDB instances.
- Collaborated with research team to build the backend service for Project True Colors. The service identifies a user's best color palettes with semantic segmentation and a custom image processing algorithm.

Visual Design and Engineering Lab at Carnegie Mellon University

Research Assistant | April 2020 – February 2021

- Implemented deep learning solutions to problems mainly within the field of additive manufacturing.
- Trained a CNN model that classifies flaws within a 3D printing process by applying transfer learning to audio.
- Developed a segmentation algorithm to create a 3D representation of 2D design components using OpenCV and PyTorch.

UC Berkeley Law School

Full-Stack Web Developer | August 2020 – May 2021

- Redesigned the Statewide Redistricting Database website on the front and back end.
- Utilized Angular, TypeScript, and Figma on the frontend; Django and MySQL on the backend.

Physics Institute of Heidelberg

Research Intern | July 2018 - August 2018

- Performed data analysis on optics experiment results
- Built an image capturing module, using Python, OpenCV, NumPy, and Pandas.

PROJECTS

JCNN

Convolutional Neural Network | Try it here!

- Efficient implementation of a CNN inference/training framework built using only Python and NumPy.
- Interactive demonstration website where a trained model classifies users' hand-drawn digits.

NeuroCars

Self-driving Car Simulation | Try it here!

- 2D self-driving car simulation which demonstrates neuroevolution in the browser.
- Agents in the simulation use neural networks to navigate tracks and iteratively evolve using the genetic algorithm.

Sorting Visualizer

Educational App | Try it here!

- Educational tool showing the individual steps of sorting algorithms in a visually pleasing way.

CashOut

Crowdsource ATM App | DVCHacks 2019 Winner | See the source code here!

- Led a team to develop a website where users post offers to exchange cash in an online money transaction.
- Flask API implementation, handlebars.js front-end templating, SQLite with SQLAlchemy.

EDUCATION

Georgia Institute of Technology

- Currently pursuing M.S. in Computer Science part-time.
- Coursework to date: Deterministic Optimization, AI Techniques for Robotics.

University of California, Berkeley

- B.S. 2022, Electrical Engineering & Computer Science
- GPA – 3.9/4.0